Introduction To Artificial Intelligence

Lecture three

Rational agent is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly.

When an agent is plunked down in an environment, it generates a sequence of actions according to the precepts it receives. This sequence of actions causes the environment to go through a sequence of states. If the sequence is desirable, then the agent has performed well. This notion of desirability is captured by a **performance measure** that evaluates any given sequence of environment states.

What is rational at any given time depends on four things:

1. The performance measure that defines the criterion of success.
2. The agent’s prior knowledge of the environment.
3. The actions that the agent can perform.
4. The agent’s percept sequence to date.

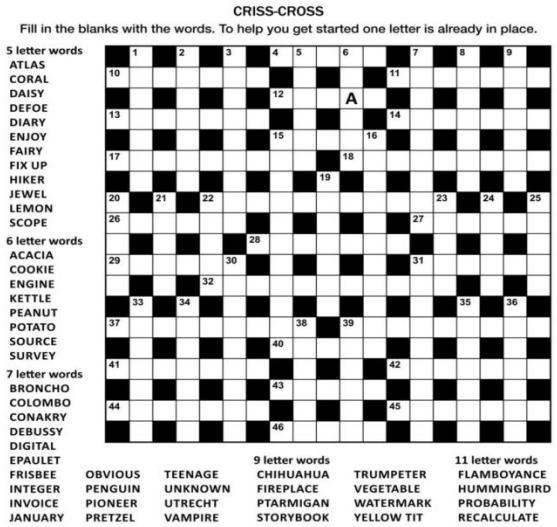
Information gathering Doing actions in order to modify future percepts and it is an important part of rationality.

Task environments which are essentially the “problems” to which rational agents are the “solutions.” In designing an agent, the first step must always be to specify the task environment. PEAS (Performance, Environment, Actuators, Sensors) description of the task environment

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

PEAS description of the task environment for an automated taxi

Part Picking Robot

Static (vs. dynamic):

- The environment is unchanged while an agent is deliberating, then, it is static. (The environment is semi dynamic if the environment itself does not change with the passage of time but the agent's performance score does)
- Static (e.g. cross-word puzzle)or 2+2=4
- Semi dynamic (e.g. chess played with clock)
- Dynamic (e.g. Automated Taxi)